

Use Case LM-1.1.1: Initialize Stage**Scope:** Launch Mission**Primary Actor:** Flight Director**Related Use Cases:** *Use Case 1: Initialize Launch,**Use Case 1.1.1.1: Initialize Engine,**Use Case 1.1.1.2: Initialize Fuel System***Stake Holders & Interests:**

- Flight Director: Wants to set up a Launch via initializing all the Launch Parameters, Systems and Sub-Systems including:
 - Rocket and Fuel System
- Mission Planner: Wants the Mission Initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The Mission System is running and ready to initialize and monitor the appropriate Systems and Sub-Systems**Post-Conditions:** The Stage is Initialized with the appropriate parameters: including everything that composes a Stage**Flow of Events:**

Flight Director	System
1. Enters the Mission Parameters File (See <i>Use Case 1:Initialize Launch</i>)	
	2. Initializes each Engine in the Stage (See <i>Use Case 1.1.1.1: Initialize Engine</i>)
	3. Initializes the Fuel System in the Stage (See <i>Use Case 1.1.1.2: Initialize Fuel System</i>)
	4. Finds the Stage Parameters File and location
	5. Goes to the Stage Parameters File location
	6. Opens the Stage Parameters file
	7. Reads the Stage Parameters file
	8. Initializes the Stage Data with the appropriate Stage data read in from the Stage Parameters File
	9. Initialization Status returned to the Flight Director

Extensions (or Alternative Flows):

- 2a If the System receives an initialization error from the Engines, then the System returns the Error to the Flight Director (See *Use Case 1.1.1.1 Initialize Engine*). The Flight Director determines the next course of action.

- 3a If the System receives an initialization error from the Fuel System, then the System returns the Error to the Flight Director (See *Use Case 1.1.1.2 Initialize Fuel System*). The Flight Director determines the next course of action.
- 4a,5a If the System cannot find the location where the Stage Parameters File is located, then the System returns the Error to the Flight Director: indicating an non-legitimate file path. The Flight Director determines the next course of action.
- 6a If the System cannot open the Stage Parameters File, then the System returns the error to the Flight Director: indicating unable to open the Stage Parameters file. The Flight Director determines the next course of action.
- 7a If the System cannot read the Stage Parameters File, then the System returns an Error to the Flight Director: indicating unable to read the Stage Parameters file. The Flight Director determines the next course of action.

Notes/Constraints:

- *a This Use Case documents the Stage Initialization, not the Stage Monitoring (See *Use Case 4.1: Monitor Initialization*) for Stage Monitoring
- *b The Next Course of Action for the Flight Director is aborting the mission or choosing another Mission Parameters File
- 9a The System indirectly alerts the Stake Holders via an upward cascade of alerts/queries with the Rocket (See *Use Case 1.1: Initialize Rocket*)

Special Requirements:

- None

Technology & Variations List:

- There could be many types of Mission Parameters Files in different formats for the components, the System will possibly need to account for all the different types of files.
- Regardless of the type of file, the System Data should follow a Standard

Frequency of Occurrence:

- Every time a Mission is planned or simulated.

Open Issues:

- None